

CHAPTER 2

Biological & Anatomical Aspects

of the Asian Elephant

Workshop Reference Chapter · Workshop on Essentials for Mortality Investigation of Asian Elephant · June 2026

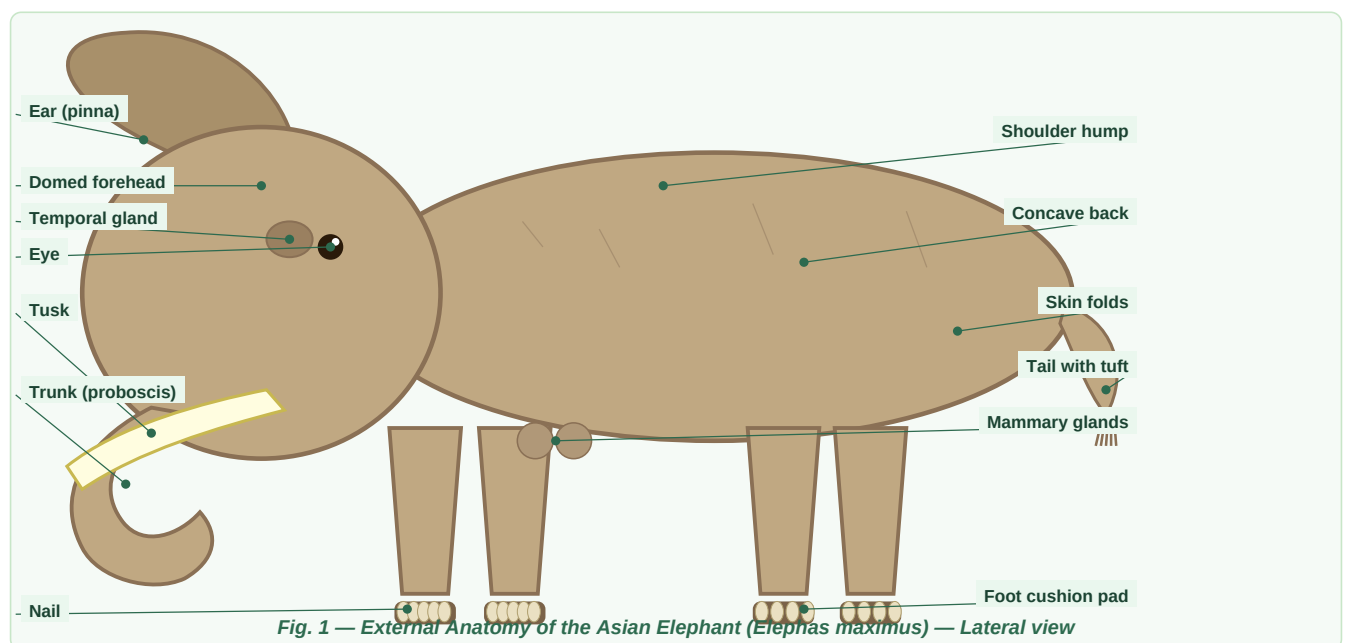
Workshop on Essentials for Mortality Investigation of Asian Elephant · Chhattisgarh Forest Department · June 2026

1. Taxonomy & Classification

The Asian elephant (*Elephas maximus*) is the largest land animal in Asia and a keystone species of tropical and subtropical forest ecosystems. It belongs to Order **Proboscidea**, the only surviving order of a once-diverse lineage of trunk-bearing mammals that included mammoths and mastodons.

Rank	Classification
Kingdom	Animalia
Phylum	Chordata
Class	Mammalia
Order	Proboscidea
Family	Elephantidae
Genus	Elephas
Species	<i>Elephas maximus</i> (Linnaeus, 1758)
Subspecies	<i>E. m. maximus</i> (Sri Lanka) · <i>E. m. indicus</i> (Indian mainland) · <i>E. m. sumatranus</i> (Sumatra) · <i>E. m. borneensis</i> (Borneo)

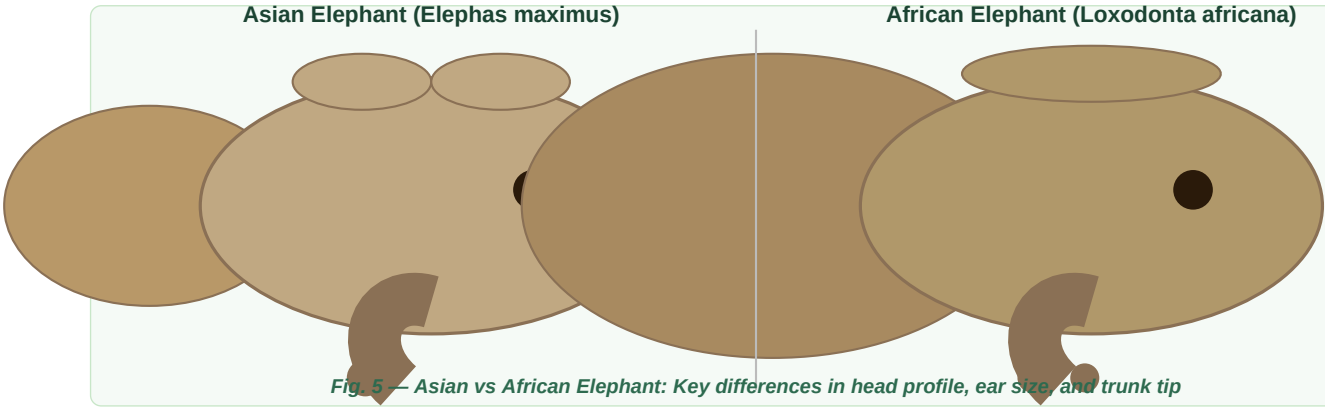
2. External Anatomy



Physical Characteristics

Parameter	Value
Shoulder height	Male: 2.5–3.5 m Female: 2.0–2.5 m
Body weight	Male: 3,500–5,000 kg Female: 2,000–3,500 kg
Body length (nose to tail)	5.5–6.5 m (male)
Skin thickness	Up to 2.5–3 cm on back/head; thinner on ears
Lifespan	60–70 years in the wild
Gestation period	~22 months (longest of any land mammal)
Inter-birth interval	3–5 years
Daily food intake	100–300 kg of vegetation
Daily water intake	100–200 litres

3. Asian vs. African Elephant — Key Differences



Feature	Asian Elephant	African Elephant
Head profile	Two-domed (divided forehead)	Single rounded dome
Ear size	Smaller (~1/3 of African)	Very large (ear = map of Africa)
Back profile	Concave / dipped	Concave / more arched
Highest body point	Shoulder	Back / mid-spine
Trunk tip fingers	1 (upper lip only)	2 (upper and lower lip)
Tusk presence	Males mainly; females often tuskless	Both sexes commonly tusked
Front foot nails	5	4–5
Hind foot nails	4	3–4
Skin texture	More wrinkled; pinkish depigmented patches common	Smoother, uniformly dark
Ear thermoregulation	Moderate (smaller pinna surface)	Very efficient (large radiator surface)

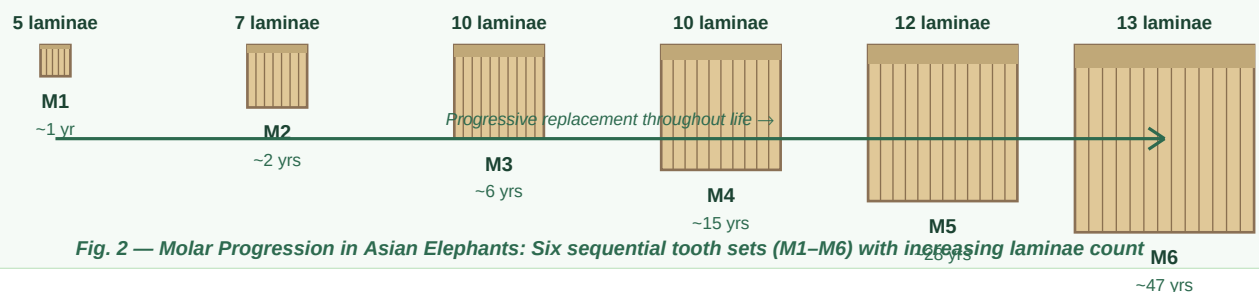
4. The Trunk (Proboscis)

The trunk is the most anatomically complex and functionally versatile organ in the animal kingdom — a **muscular hydrostat** (like the human tongue) containing approximately **100,000 individual muscle units** but

no bone whatsoever. It is a fusion of the elongated nose and upper lip.

Function	Details
Breathing	Primary airway; can breathe under water by using as a snorkel
Drinking	Sucks up 8–10 L per draw; transfers to mouth or sprays onto body
Bathing & cooling	Sprays water or mud over back and sides for evaporative cooling
Olfaction	Raised to detect scents at height; most important sensory function (up to 2,000 olfactory receptor genes)
Grasping & foraging	1 "finger" at tip (Asian) allows precise picking of leaves, fruits, and grass; can uproot trees with trunk-and-foot leverage
Vocalisation	Produces trumpeting calls (alarm, greeting, excitement); low-frequency rumbles for infrasound communication
Social bonding	Touching, intertwining, and exploring other elephants; used in greeting and consolation
Defence & combat	Striking, throwing, and pushing; males use in musth conflicts
Digging	Excavates water holes in dry riverbeds; extracts salt licks
Disease sampling	Trunk wash: key clinical sample for TB and EEHV diagnosis

5. Tusks, Teeth & Dentition



Tusks

- Tusks are **modified second upper incisor teeth** — they grow continuously throughout the elephant's life from deep sockets in the premaxillary bone.
- Composition:** solid dentine (ivory) with a small pulp cavity at the root; the outer surface has a thin enamel cap that wears off early in life.
- In Asian elephants, **only adult males** normally carry large visible tusks. Females usually have small **tushes** (rudimentary tusks) or none at all.
- Males of the **Sri Lankan subspecies** (*E. m. maximus*) are frequently tuskless — a genetically and selectively driven trait.
- Permanent tusks **protrude beyond the lips at ~30 months** and are used for digging, debarking trees, combat, and as resting props for the trunk.

Molar Dentition

Elephants are **unique among mammals** in producing six successive sets of molar teeth (M1–M6) that erupt and move forward (mesiodentally) along the jaw during life. Each molar is larger than the previous and contains more **laminae** (transverse enamel ridges that provide grinding surface). When M6 is exhausted — typically at 55–65 years — the animal can no longer chew adequately and eventually dies of malnutrition.

6. Ears & Thermoregulation

The elephant ear is a masterpiece of biological engineering — a **vascularised thermal radiator**. A dense network of superficial blood vessels runs through the ear's thin skin, and rapid flapping creates a convective airflow that dissipates heat from the blood before it returns to the core.

Aspect	Details
Vascular network	Dense capillary plexus visible through thin ear skin — blood cooled by 3–9 °C before returning to core circulation.
Ear flapping	Slow rhythmic flapping (1–3 times/min at rest; faster in heat) acts as a bio-fan, dramatically increasing convective heat loss.
Surface area	Asian elephant ear $\approx$ 0.3–0.5 m ² per ear; African ear $\approx$ 1.5–2.0 m ² — this explains why African elephants tolerate greater heat.
Communication	Ear posture signals mood: forward/spread = alert/threatening; flat against neck = submission or contentment.
Infrasound coupling	Ground vibrations from infrasound calls may be sensed through the feet AND detected via the jaw/skull; ear anatomy plays a secondary role.

7. Skin

Property	Detail
Thickness	Up to 2.5–3 cm on the back and head; much thinner (1–2 mm) on the ears
Colour	Dark grey-brown; depigmented pink-white patches common around ears, trunk, and face (especially in Asian elephants)
Texture	Heavily wrinkled and folded — wrinkles increase surface area for evaporative water loss; reduce surface temperature
Sweat glands	Absent except for small glands between the toes; transepidermal water loss through permeable skin is the main mechanism
Thermoregulation	Skin permeability increases in heat — effective when drinking water available; mud and water bathing critical
Health indicators	Cracked, dry, or peeling skin: dehydration or nutritional deficiency. Pox lesions, skin tumours, cutaneous filariasis all externally visible

8. Feet & Locomotion

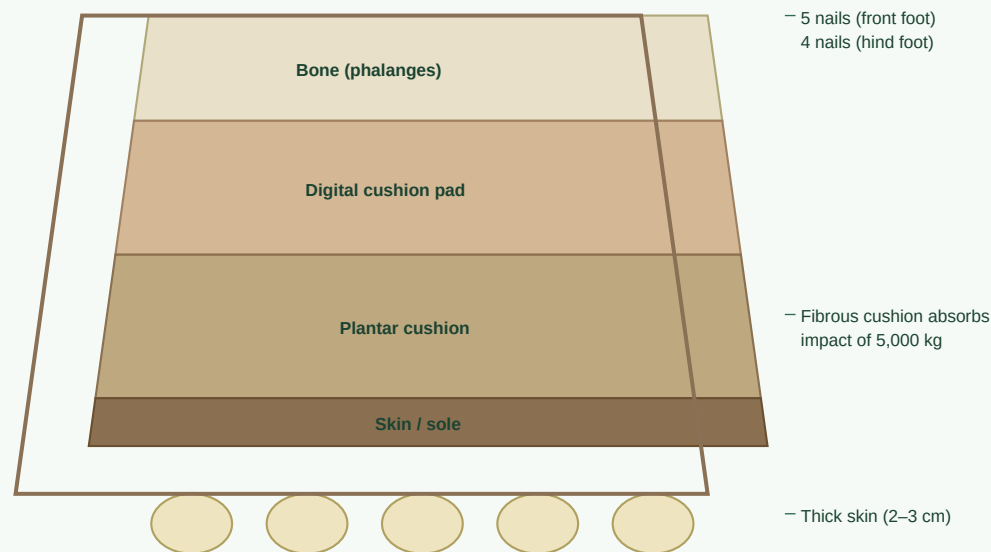


Fig. 3 — Cross-Section of the Asian Elephant Foot (showing cushion pad layers and nails)

- **Semi-digitigrade posture** — elephants walk on the tips of their digits, with a large fibrous cushion pad (like a built-in heel pad) absorbing impact.
- The cushion pad acts as a **hydraulic shock absorber**, distributing the 3,500–5,000 kg body weight across the foot to reduce pressure per cm².
- Despite enormous weight, elephants walk **almost silently** — the cushion pad muffles footsteps, essential for stealthy forest movement.
- **Maximum speed:** ~25 km/h (elephants do not truly gallop; all four feet never leave the ground simultaneously).
- **Asian elephant nails:** 5 on the front foot, 4 on the hind foot. Overgrown, cracked nails are a common captive management problem.
- Footpad health is critical — infections or injuries in foot pads can be life-threatening for captive elephants.

9. Skeletal System

Structure	Feature
Cervical vertebrae	7 (same as all mammals)
Rib pairs	20 pairs (most land mammals have 12–13)
Lumbar vertebrae	3–4 (fewer than most mammals; aids weight-bearing)
Bone density	Pachyosteosclerosis — bones are extremely dense and compact, providing structural support for massive weight
Femur & humerus	Vertically positioned (columnar) — unlike most quadrupeds; entire leg acts as a column, not a pendulum
Skull	Large but surprisingly light — numerous air-filled diploe (cancellous bone chambers) reduce skull weight
Foot bones	All five digits present; digit bones (phalanges) short; sesamoid bones within the cushion pad
Tusk attachment	Deep root in premaxillary bone; periodontal ligament allows slight flexibility during use

10. Digestive System

100–300 kg food/day · 100–200 L water/day · ~40% digestive efficiency · 24–48 hr GI transit

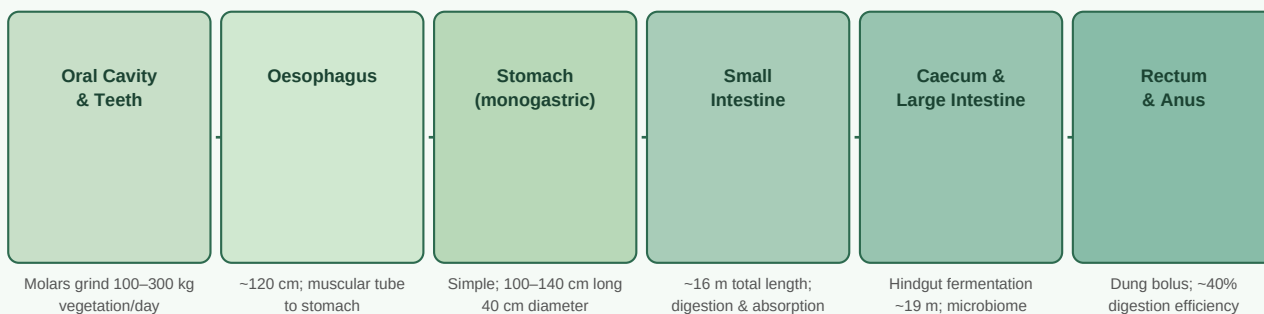


Fig. 4 — Digestive System of the Asian Elephant (Hindgut Fermenter — Monogastric)

- **Monogastric (simple-stomached)** — unlike ruminants, elephants have a single-chambered stomach (~100–140 cm long, ~40 cm diameter).
- **Hindgut fermentation** — microbial breakdown of cellulose occurs primarily in the enlarged caecum and colon (not in the stomach/small intestine).
- **Relatively low digestive efficiency (~40%)** — compensated by the enormous volume consumed (100–300 kg/day).
- **Total intestinal length** can reach ~35 metres (stomach + small + large intestine + caecum + rectum).
- **GI transit time:** 24–48 hours — undigested seeds dispersed over large distances (elephants are keystone seed dispersers).
- **Clinical relevance:** sudden feed changes → gut microbiome disruption → colic, diarrhoea, or increased toxin production.

11. Reproductive System

Feature	Detail
Mammary glands	Located between the forelimbs (pectoral position) — unique among large mammals; two glands with one teat each
Oestrous cycle	~16 weeks; ovulation associated with two LH surges
Gestation	~22 months (longest of any land mammal)
Age at first calving	Females: ~14 years; males not usually successful until >20 years
Inter-birth interval	3–5 years (limited by extended lactation)
Sexual maturity (male)	~15 years; socially mature at 25–30 years
Musth	Annual period of heightened testosterone (10–100× normal) in adult males; temporal gland secretion, urine dribbling, aggressive behaviour; lasts days to months
Temporal gland	Paired exocrine glands between eye and ear; active during musth and in females during oestrus; examined at necropsy
Testes	Internally located (abdominal); not externally visible
Penis	S-shaped when erect; males urinate backward
Placentation	Zonary haemochorial; relatively large placenta

12. Brain, Senses & Communication

Brain

- The Asian elephant brain weighs **~4–5 kg** — the largest brain of any land mammal in absolute terms.
- Highly convoluted cortex; well-developed **hippocampus** (memory) and **temporal lobe** (social cognition).
- Demonstrated capabilities include: **mirror self-recognition**, **tool use**, empathy, altruism, and long-term memory (>20 years).
- Encephalisation Quotient (EQ) is lower than primates but higher than most other large mammals.

Sense	Mechanism & Notes	Capability
Olfaction (smell)	Most dominant sense. >40% of genome dedicated to olfactory receptors — more than any other animal studied. Trunk raised to sample scent plumes at height.	Exceptional (>2,000 OR genes)
Infrasound hearing	Produce and detect calls at 14–35 Hz (below human hearing range). Seismic vibrations also sensed through feet and jaw.	Communication up to 10 km
Tactile sense	Trunk tip is exquisitely sensitive (Meissner's corpuscles). Sensitive skin on feet detects ground vibrations.	Excellent
Vision	Relatively poor compared to other senses. Dichromatic colour vision (blue-yellow; red appears as green). Better in dim light than bright sun.	Limited; poor in bright light
Taste	Well-developed; use to select preferred food plants and detect contaminants or toxic plants.	Good

Key Takeaways

Chapter Summary — Biological & Anatomical Aspects of the Asian Elephant

1. Asian elephants (*Elephas maximus*) are identified by their double-domed head, smaller ears, concave back, and single trunk-tip finger — key distinctions from the African elephant.
2. The trunk is a boneless, 100,000-muscle organ critical for survival; trunk wash is the priority clinical sample for EEHV and TB diagnosis.
3. Six successive molar sets (M1–M6) allow age estimation; exhaustion of M6 at 55–65 years leads to eventual death from inability to chew.
4. The foot cushion pad silently absorbs enormous weight; nails and pads are critical health indicators in captive animals.
5. Hindgut fermentation in the caecum/colon with low (~40%) efficiency is compensated by enormous daily food intake (100–300 kg).
6. 22-month gestation, 3–5 year inter-birth interval, and slow sexual maturity make the population extremely sensitive to adult mortality.
7. Olfaction (>2,000 olfactory receptor genes) and infrasound communication (detectable up to 10 km) are the dominant senses.